Hasti Barghararianyan

Location: France | Open to relocation
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EDUCATION

M.Sc. Public Health, Track: Comparative Effectiveness Research Oct 2025 – Jul 2026
Université Paris Cité, Paris, France

- **Relevant coursework:** Statistics, Observational Studies, Randomized Controlled Trials, Target Trial Emulation, Systematic Reviews and Meta-Analysis, Network Meta-Analysis, Personalized Medicine.

M.Sc. Computer Science (Year 1), Track: Machine Learning and Data Mining Sep 2024 – Jul 2025
Université Jean Monnet, Saint-Étienne, France

- **Relevant coursework:** Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Analysis, Optimization, Algorithms and Programming, Complexity Theory.

B.Sc. Mathematics, Minor in Computer Science Sep 2019 – Jun 2024
Tehran Polytechnic (Amirkabir University of Technology), Tehran, Iran

- **Relevant coursework:** Mathematical Analysis, Numerical Analysis, Linear Algebra, Probability, Graph Theory, Optimization, Databases, Operating Systems, Computer Networks, Computer Simulation, Artificial Intelligence.

RESEARCH AND PROFESSIONAL EXPERIENCE

Master's Thesis – Public Health Research Intern, OPPaLE team, CRESS (INSERM) Feb 2026 – Jul 2026
Paris, France. Supervisors: Dr. Marion Lecorguillé and Prof. Marie-Aline Charles

Project: Smoking during pregnancy and child long-term health using an outcome-wide approach in the ELFE cohort

- Examined the association between maternal smoking during pregnancy and four domains of child health in the ELFE birth cohort (N ≈ 18,000): respiratory disorders, neurodevelopmental disorders (cognitive development and mental health), sleep quality, and obesity.
- Applied causal inference and statistical methods (DAG-based confounder selection and regression adjustment) and handled missing data with machine learning-based multiple imputation; full pipeline in R.
- Presented to the collective lab (Jun 2026)
- Abstract accepted at the DOHaD conference (Munich, Sep 2026).

Research Assistant, METHODS team, CRESS (INSERM) Nov 2025 – Apr 2026
Paris, France

- Contributed to two systematic reviews, performing title/abstract screening (≈ 1,500 records) and supporting literature search strategies.
- Supported an AI-driven healthcare research project in two phases: (1) manually validated AI-generated records (≈ 200 records), each with multiple outcomes, confirming the correct version of every entry; (2) executed search strategies across bibliographic databases, including PubMed and the Cochrane Library.

Machine Learning Research Intern, Hubert Curien Laboratory Mar 2025 – Jul 2025
Saint-Étienne, France. Supervisors: Prof. Antoine Gourru, Prof. Jairo Cugliari, and Prof. Eduardo Brandao

Project: MOLE – Model Optimization from Learning Examples

- Applied meta-learning to optimize the parameters of neural networks; implementation in Python.
- Conducted an extensive literature review on meta-learning and generative models, in particular diffusion models.
- Presented the project at the laboratory Interns' Day (Jul 2025).

Research Collaborator, Remote Jan 2025 – Apr 2025
Remote, based in France. Collaboration with Prof. Babak Khoshnood

Project: Systematic review and meta-analysis: Vitamin B intake and CHD risk

- Screened articles (≈ 500 records).
- Poster presented at the AEPC Annual Meeting (Hamburg, May 2025).

Bachelor's Thesis Researcher, Tehran Polytechnic Oct 2022 – Oct 2023
Tehran, Iran. Supervisor: Prof. Farnaz Hooshmand Khaligh

Project: Algorithm based on clustering approach for asset portfolio selection

- Researched different approaches to finding profitable investment opportunities.
- Applied K-means clustering to stock data of 5 companies to maximize investment returns.

Programming Teacher (Scratch, Python), Yasan Academy
Tehran, Iran. Jun 2022 – Aug 2022

- Taught two classes of 15 children aged 6–15; received positive feedback from parents.

PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

- Barghararian H., Charles M-A., Lecorguillé M. *Smoking during pregnancy and child long-term health using an outcome-wide approach in the ELFE cohort.* Manuscript in preparation for submission, 2026.
- Abstract accepted, DOHaD conference, Munich, Germany, Sep 2026 (related to the manuscript above).
- Lab presentation, OPPALE team, CRESS, Paris, France, Jun 2026 (related to the manuscript above).
- Interns' Day presentation, Hubert Curien Laboratory, Saint-Étienne, France, Jul 2025: MOLE – Model Optimization from Learning Examples.
- Poster, AEPC Annual Meeting, Hamburg, Germany, May 2025: Systematic review and meta-analysis: Vitamin B intake and CHD risk.

SKILLS

Programming Languages	R, Python, SQL, C, C++, MATLAB, Prolog
Statistical & Causal Methods	DAGitty, Regression & Causal Modeling, Multiple Imputation
Data Science & Machine Learning	NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow
Systematic Review & Meta-Analysis	SK Platform, Zotero, Rayyan, Covidence, PRISMA, RoB, GRADE
Development & Reproducibility	Git, Conda, pip, LaTeX, R Markdown
Optimization	PuLP, Pyomo, GAMS, Ipopt
Web Technologies	WordPress, HTML, CSS
Operating Systems	Linux (Ubuntu), Windows

LANGUAGES

English: Fluent, C1 (TOEFL iBT 95: R 24, L 22, S 25, W 24) | **French:** A1, currently improving | **Azerbaijani:** A1
Persian: Native

EDITORIAL AND LEADERSHIP ACTIVITIES

Editor-in-Chief, Math Section, Halgheh (student journal) Jul 2021 – Mar 2023
Department of Mathematics & Computer Science, Tehran Polytechnic, Tehran, Iran

- Coordinated cross-functional teams (authors, editors, design, media, and interviews), managing deadlines and ensuring smooth collaboration between groups.
- Planned each volume with the head manager, recruited talented contributors, and liaised with the relevant association regarding funding.
- Helped the journal grow rapidly and gain recognition among universities and schools.
- Mentored the incoming team to ensure a smooth handover.
- Developed leadership, teamwork, and consistency skills.

Scientific Author, Math Section, Halgheh Oct 2020 – Jul 2021
Department of Mathematics & Computer Science, Tehran Polytechnic, Tehran, Iran

Authored articles in the following volumes:

- Volume 1 | Conway Knot: Graduate Student, Lisa Piccirillo, Solved Decades-Old Conway Knot Problem
- Volume 2 | Topology: What is Topology?
- Volume 3 | Fractal Geometry: A short introduction to Fractal Geometry
- Volume 4 | Cantor Set: How is Cantor set made and What are its features?

EVENTS ATTENDED

Attendee, Healthcare Week Luxembourg Sep 2026
University of Luxembourg, Luxembourg

Attendee, scientific event «Quantum Breakfast: Quantum, AI, and Health» Sep 2026
University of Luxembourg, Luxembourg

Attendee, scientific lecture «The impact of alcohol on the global burden of cancer: evidence and estimates from epidemiological studies» Sep 2026
Department of Precision Health, LIH, Strassen, Luxembourg

Attendee, scientific symposium «How AI and LLM are transforming research in Public Health?» Dec 2025
CRESS, Université Paris Cité, Paris, France

Attendee, 13th SLEIGHT Science Event Jan 2025
Manufacture Campus, Université Jean Monnet, Saint-Étienne, France